TECHNICAL DESIGN DOCUMENT

FUTURE ENGINEERS

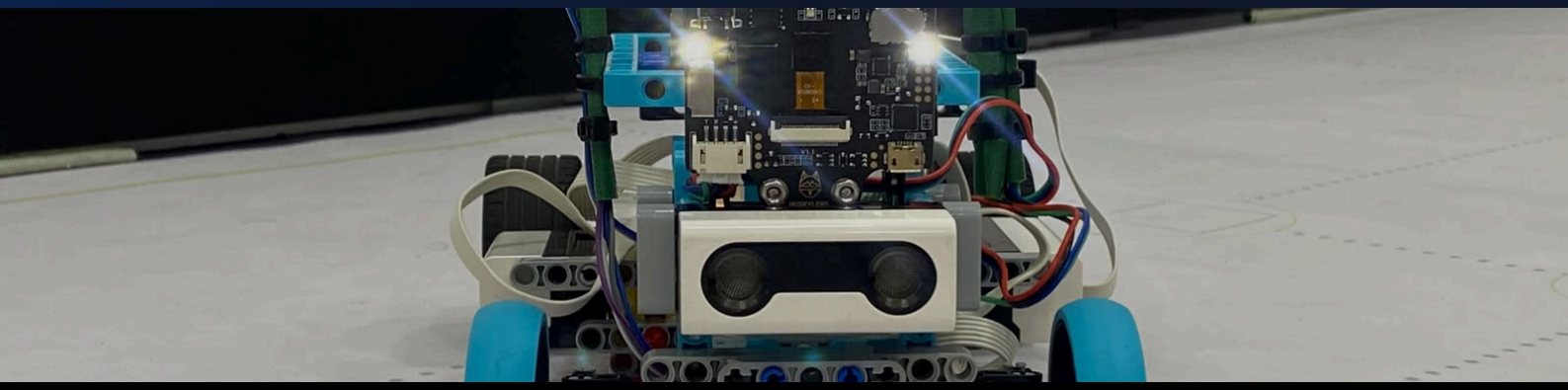
A robot that follows the wall to the centimeter, reads colored signs with an AI eye, and decides with a state machine.

01 · MECHANICS | 02 · POWER & SENSORS | 03 · SOFTWARE | 04 · SYSTEMS | 05 · REPRODUCIBILITY

AQSA SAUDI FORCE

WRO-FE-2026 · REV B · 20 SHEETS

"No decision without evidence — across twenty sheets we documented every engineering choice."



00 · SHEET 02

Team & Coach

Roles, the rubric map, and the working method this entire document is built on

We are a team that works with a full engineering mindset: clear roles, decisions justified by numbers, and continuous documentation of every experiment. We divided our work across the five official rubric criteria and set the maximum level (**L6**) as our target on each — this expanded document (20 sheets) is the field evidence of that commitment.

M

Maher Al-Harbi

MECHANICAL DESIGN & STEERING

Leads the design of the modular SPIKE chassis and the steering system; runs steering-angle calibration sessions and measures deviation after every change.

B

Bader Bukhari

SOFTWARE — PYBRICKS & STATE MACHINE

Builds the PID controller and the state machine in Pybricks, and tunes the gains until tracking settles without oscillation.

C

Mohammed Basuny

COACH

Guides us without driving for us: asks the hard questions, makes sure every engineering decision is truly ours, and holds us to documentation first.

CRITERION 01	CRITERION 02	CRITERION 03	CRITERION 04	CRITERION 05
Mechanical	Sensors & Power	Software	Systems	Reproducibility
L6	L6	L6	L6	L6

We log

Every test run gets a line in our log: conditions, result, and the engineering note.

We measure

No decision without a number: lap time, lateral deviation, vision success rate.

We document

Every change goes into the repository with its reason — the team's collective memory.

00 · SHEET 03

Document Map & Reading Guide

Where to find every rubric requirement across the twenty sheets

The document is designed to be read as one continuous path: from raw material to software behavior to the open guide. A judge who prefers selective auditing gets a direct map — each of the five rubric criteria owns dedicated sheets flagged by its color tab, and every engineering claim in this file is backed by a number, an image, or a table.

SHEET	SECTION	CRITERION	HIGHLIGHTS
P01	Cover	—	Team identity, document scope, five criterion badges
P02	Team & Coach	—	Roles, methodology, rubric map
P03	Document Map	—	This map — full requirement tracking
P04	1.1 Chassis & Materials	RUBRIC 01	Strength-to-weight ratio, torsional rigidity, center of gravity
P05	1.2 Ackermann Steering	RUBRIC 01	Independent steering, home alignment, absolute encoder
P06	1.3 Drivetrain & Traction	RUBRIC 01	Weight over the drive axle, **1:3 speed gears**, grip force, stability
P07	1.4 Manufacturing Iterations	RUBRIC 01	V1→V2: bracing, sensor mounts, cable management
P08	2.1 Sensor Architecture	RUBRIC 02	3× ultrasonic + HuskyLens + gyro + color
P09	2.2 Power Budget	RUBRIC 02	Full consumption table, capacity, wiring diagram
P10	2.3 Calibration & Failure	RUBRIC 02	EMI, self-calibration, sensor fusion on failure
P11	3.1 Development Environment	RUBRIC 03	Pybricks, module architecture, execution loop
P12	3.2 PID Controller	RUBRIC 03	Equation, parameter roles, tuning log, response curve
P13	3.3 Finite State Machine	RUBRIC 03	Full diagram, transition table, timing loop
P14	3.4 Edge Cases	RUBRIC 03	Vision loss, wall gap, recovery pattern & watchdog
P15	4.1 Engineering Trade-offs	RUBRIC 04	Why X not Y — number-backed cards + **1:3 gear** trade-off
P16	4.2 Iterations & Results	RUBRIC 04	V1→V2 log and the official field performance table
P17	4.3 Safety & Risk	RUBRIC 04	Electrical, mechanical, operational + pre-run checklist
P18	5.1 Repository & Quick Start	RUBRIC 05	Full GitHub structure + rebuild in 55 minutes
P19	5.2 Quality & BOM	RUBRIC 05	Checklist, bill of materials, release log
P20	Closing & Commitment	—	L6 commitment summary and document revision log

Logical sequence

Each section builds on the previous: mechanics carry the sensors, sensors feed the algorithms.

Claim ≠ evidence

No judgment in this file without a measurement, an image or a table behind it.

Selective reading

Each sheet's colored tab reveals its rubric criterion instantly — no flipping required.

RUBRIC 01 · MECHANICAL · L6

Chassis Architecture & Design

Material selection, the modular approach, and dynamic balance

We adopted a hybrid structural system built on **LEGO® Technic / SPIKE™ Prime high-impact ABS composites**, for their rare balance of rigidity, weight, and near-instant reconfiguration between test sessions — a decision we justify with numbers on the trade-offs sheet (P15). The chassis is not a container for electronics but an active performance element: its rigidity protects sensor accuracy, its low center of gravity converts motor force into useful motion instead of wasted slip, and its reconfigurability turned every test session into an improvement opportunity.

MATERIAL	ABS COMPOSITE	CHASSIS LAYERS	1 LAYER	CENTER OF GRAVITY	LOW CoG

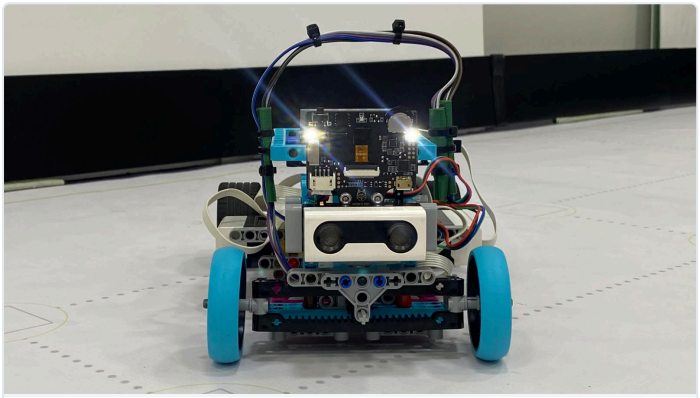


FIG 1.1-A · Chassis front view — SPIKE hub, drive motor, and HuskyLens mounted forward

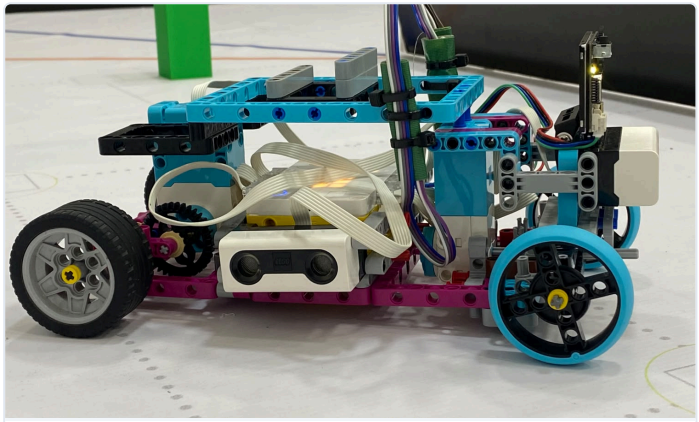


FIG 1.1-B · Field deployment — modular chassis on the WRO 2026 arena

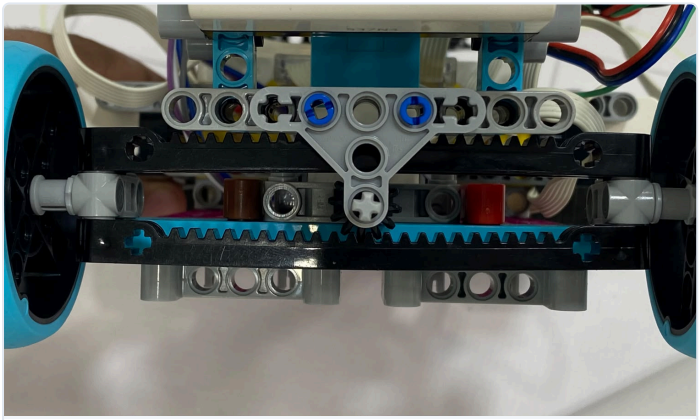


FIG 1.1-C · Underside — Ackermann steering linkage and drive axle visible

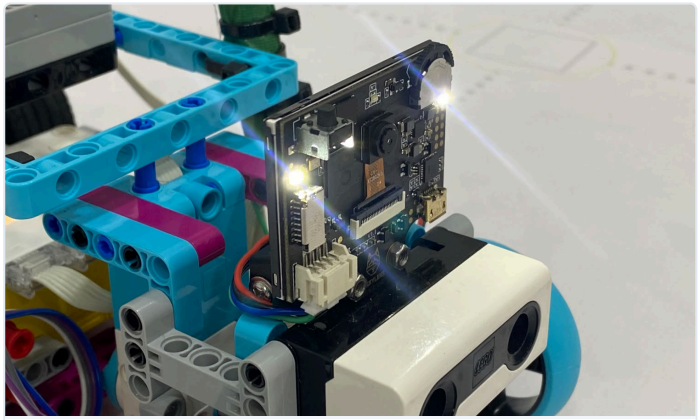


FIG 1.1-D · Top view — HuskyLens camera mount and cable routing

- **Strength-to-weight ratio:** Light, high-stiffness polymers reduce the robot's mass moment of inertia, improving angular acceleration without sacrificing chassis rigidity.
- **Torsional rigidity:** Cross-bracing with multi-point beams prevents structural deflection under lateral forces during fast cornering.
- **Low center of gravity:** Hub and motors sit at the lowest possible point inside the chassis — higher grip, less slip in hard maneuvers, no rollover.
- **Vibration damping:** The stacked chassis naturally absorbs shocks and micro-vibrations from tire contact, protecting sensors from mechanical noise.

Engineering summary: The chassis is not a container for electronics but an active performance element — its rigidity protects sensor accuracy, its low center of gravity converts motor force into useful motion instead of wasted slip, and its reconfigurability turned every test session into an improvement opportunity.

RUBRIC 01 · MECHANICAL · L6

Steering System — Ackermann Geometry

A true RC-car style steering axle with proportional angles and instant response

The front axle was designed around a **dedicated servo motor driving the wheel angles per Ackermann steering geometry** — the same principle as RC cars: each wheel pivots around a shared turn center, reducing tire drag and preventing side slip in hard maneuvers, instead of a single uniform angle that scrubs the tires sideways. This geometry is the essence of the Future Engineers category — it mirrors how real future cars steer.

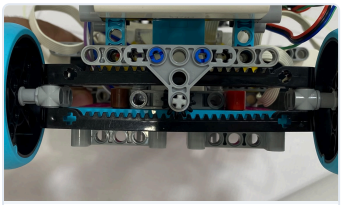


FIG 1.2-A · Steering linkage — servo, tie rods, and Ackermann arms

- **Ackermann geometry:** Each wheel's axle extension passes through one turn center, keeping angles proportional and drag minimal.
- **Home alignment:** A routine re-centers the steering angle to zero at every boot — every run starts with a proven straight heading.
- **Absolute encoder:** Real-time angular position reporting gives fast response and instant correction of any deviation.
- **Dual steering limits:** A software limit plus a mechanical limit prevent abrupt changes that would break tire contact with the ground.
- **Max steer angle:** ±40° per front wheel — enough for the tightest corner in the official arena.

FRONT AXLE	Steered, Ackermann geometry	STEER ANGLE	±40°	STEER MOTOR	SPIKE Medium Motor

Calibration & response: Boot routine zeroes the servo, the absolute encoder reports position every cycle, and software + mechanical limits protect tire contact through every maneuver.

Why Ackermann, not differential? Higher numerical stability — differential drive deviated in our measurements at high speed, while Ackermann held the target line. Less drag in corners means lower friction and precise repetition of the same arc.

RUBRIC 01 · MECHANICAL · L6

Drivetrain & Traction — Dynamic Balance

Weight over the drive axle, wheel selection, and motion stability

Our motion layout follows one simple rule: **force is delivered where the weight is** — two rear drive wheels carry the robot's heaviest elements (hub and battery) directly above them, converting motor torque into effective grip instead of spinning wheels. A **1:3 speed gear train** is installed between the drive motor and the rear wheels, multiplying the wheel speed threefold (big Z36 input drives small Z12 output) and giving the robot higher top speed on straights.

1 : 3

Speed multiplier ×3 — gear train installed between drive motor and rear wheels. Bevel gear set with Z36 input driving Z12 output (big → small) gives 3× output speed and ⅓ output torque: higher top speed on straights, faster lap times.

DRIVE	2 × rear wheels	STEERING	Front, Ackermann	WEIGHT DISTRIBUTION	Over drive axle
CENTER OF GRAVITY	Lowest in chassis	GEAR RATIO	1 : 3 (speed ×3)	WHEEL DIAMETER	56 mm

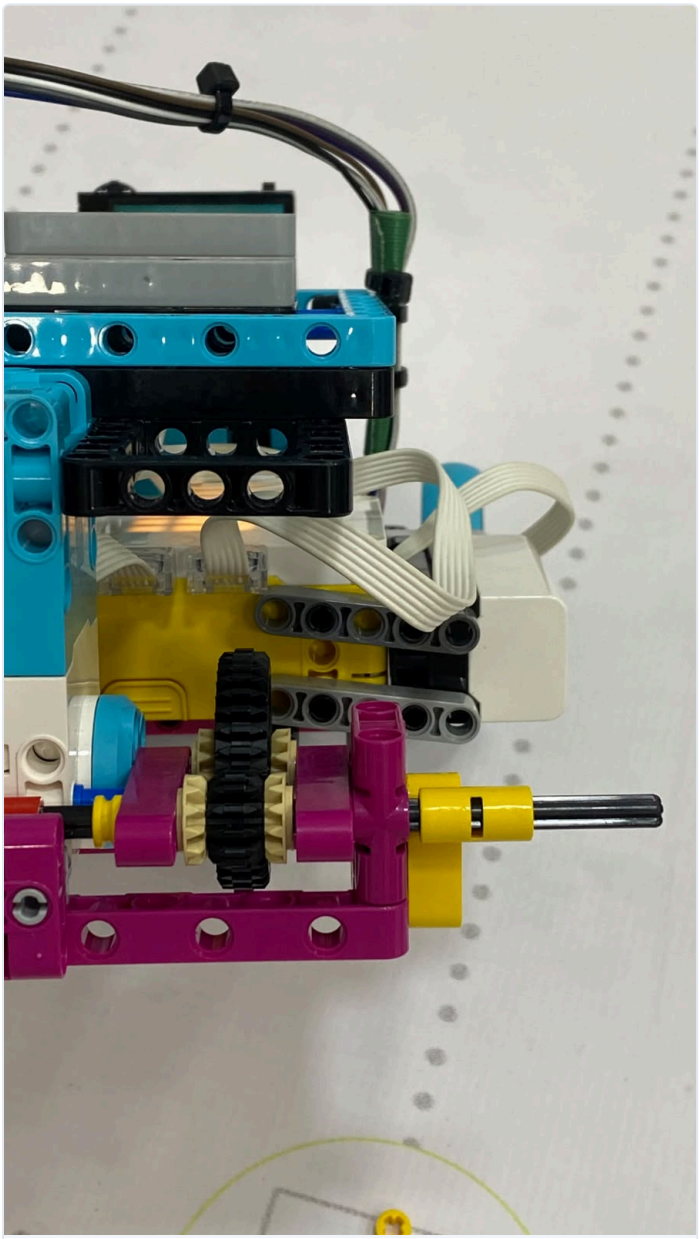


FIG 1.3-A · Installed 1:3 gear train — bevel gears between drive motor and rear axle

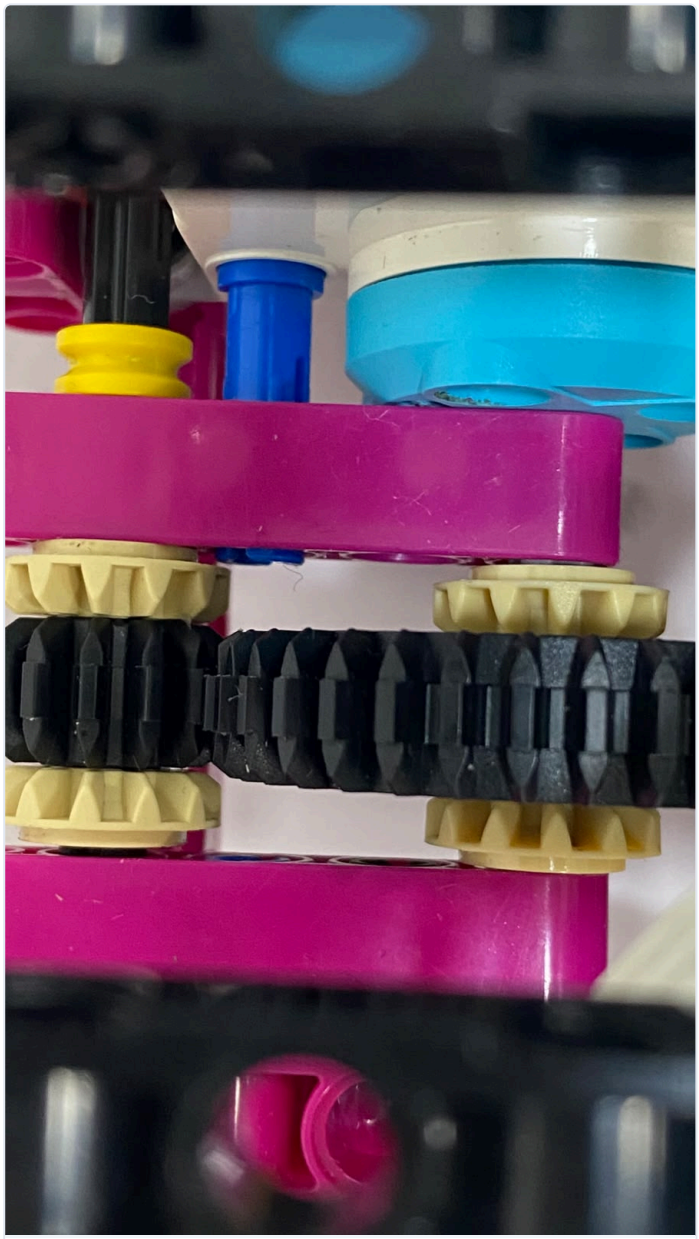


FIG 1.3-B · Gear mesh macro — Z36 input driving Z12 output, threefold speed multiplication

RUBRIC 01 · MECHANICAL · L6

Manufacturing Decisions & Design Iterations

From V1 to V2: bracing, sensor mounts, and cable management

Every field defect gets a **"ticket"**: description, quantitative measurement of the problem, applied fix, then re-measurement to confirm — the same mindset industry teams use. Without the V1 rounds, V2 would never have been born. The modular polymer system allowed weight redistribution and axle-angle experiments at near-zero time cost between field test sessions.

V1 — First prototype

- Unwanted mechanical flex at the front-axle joint during sudden steering — reflected directly in sensor reading stability.
- Drivetrain ran at motor speed only — no speed multiplication, full motor speed delivered directly to the wheels.
- Sensor mounts drifted under vibration — readings contaminated by micro-motion.

V2 — FINAL

- Bracing + additional orthogonal mounting points → higher overall rigidity and settled sensor stability.
- **1:3 speed gear train installed** between drive motor and rear axle — threefold speed multiplication (Z36 → Z12), higher top speed on straights.
- Purpose-built integrated mounts guarantee zero vibration and no viewing-angle drift while moving.

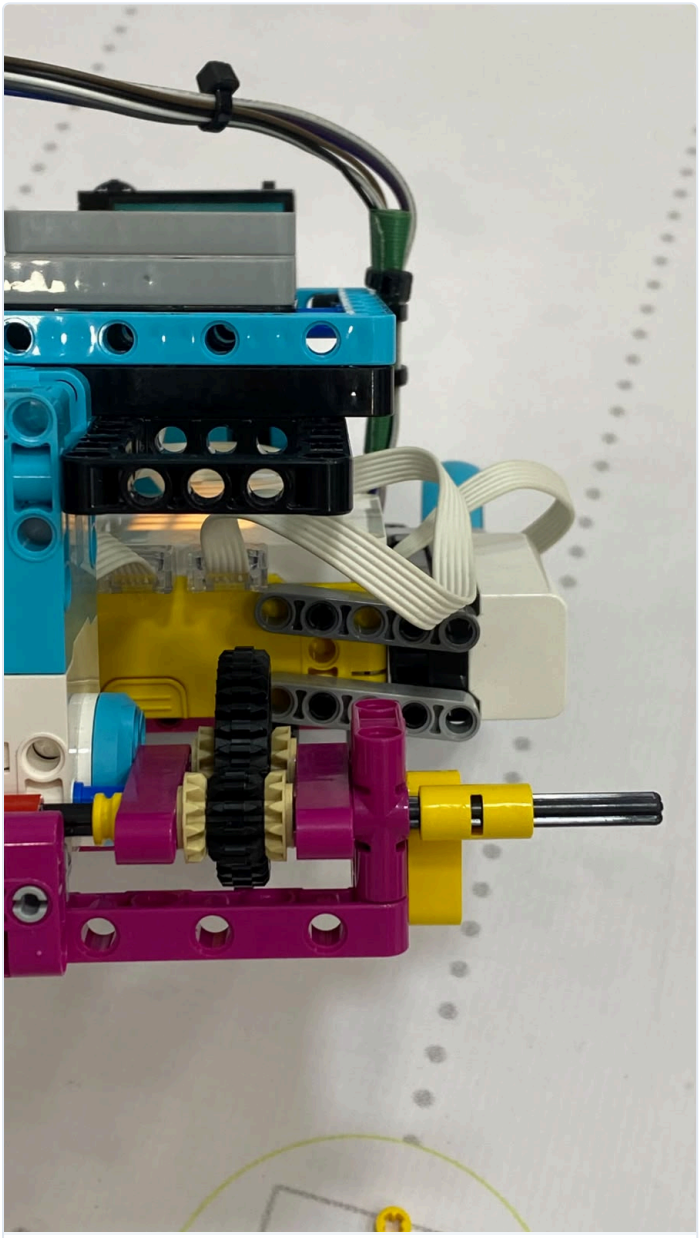


FIG 1.4-A · V2 final — bevel gear set delivering 1:3 speed multiplication (Z36 → Z12)

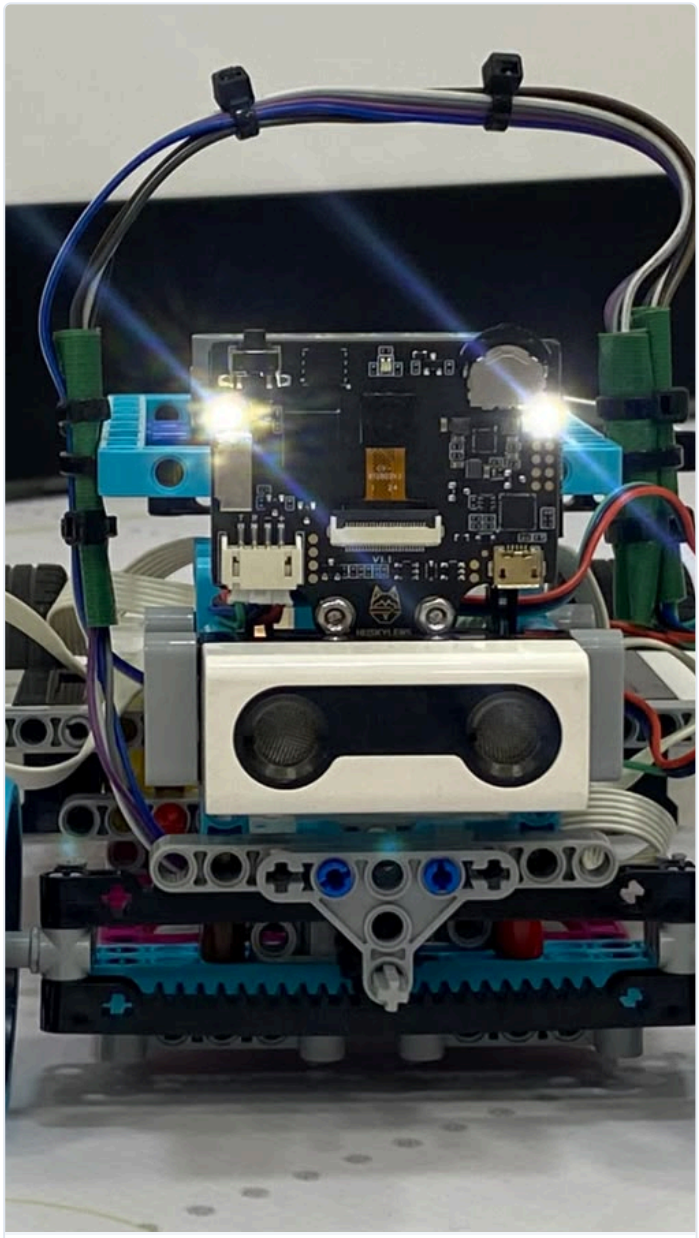


FIG 1.4-B · V2 final assembly — bracing, integrated mounts, internal cable channels

RUBRIC 02 · SENSORS · L6

Sensor Architecture & Spatial Distribution

A hybrid sensing suite delivering full environmental awareness

We integrated **five complementary perception systems**: three ultrasonic distance sensors covering front and both sides, an AI camera for colored signs, a six-axis gyro for heading, and a downward color sensor for floor lines — each one covers another's blind spot. Ultrasonics see distance but not color, HuskyLens sees color but not distance, and the gyro holds heading when both fail — this integration is the essence of "full environmental awareness".

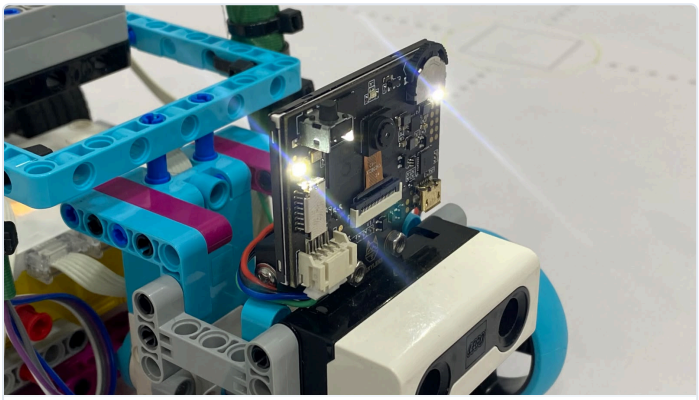


FIG 2.1-A · HuskyLens AI camera — mounted on the front face for color sign recognition

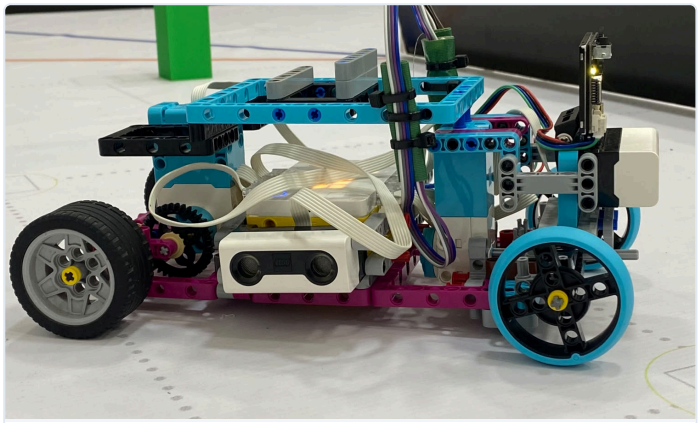


FIG 2.1-B · All five sensors live — robot running on the WRO 2026 field

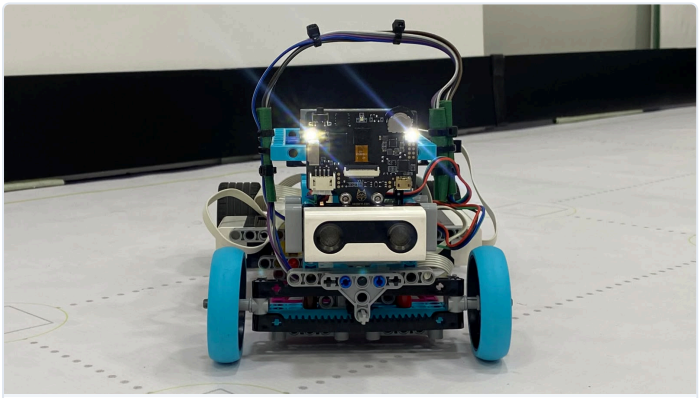


FIG 2.1-C · Front sensor array — HuskyLens + front ultrasonic + cable routing

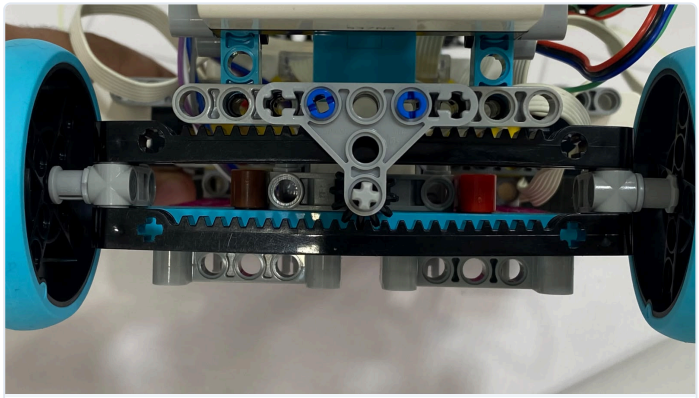


FIG 2.1-D · Side ultrasonic placement — clear view of the wall-following plane

- **Front ultrasonic:** Obstacle detection up to **100 cm** with speed control and a safe stopping distance.
- **Left & right ultrasonics:** Constant **20 cm** wall parallelism — the foundation of continuous tracking.
- **HuskyLens AI:** On-board recognition of colored signs without loading the main hub — green = left, red = right.
- **6-axis gyro + downward color sensor:** Yaw/Pitch/Roll for angular stability, plus floor line and field boundary detection.

Why this combination? Each sensor covers its sibling's blindness — ultrasonics see distance but not color, HuskyLens sees color but not distance, and the gyro holds heading when both fail. This integration is the essence of "full environmental awareness".

RUBRIC 02 · POWER · L6

Power Budget & Energy Distribution

Full calculation of nominal current, stall current, and power per component

We calculated the maximum and nominal draw of every component to guarantee the robot **never suffers a voltage drop** during hard maneuvers — the table below is the energy contract all powering decisions were built on. The integrated lithium battery (2100 mAh) delivers over **120 minutes of continuous operation** under maximum load — double the competition's run requirements, with a wide safety margin.

COMPONENT	V	NOMINAL MA	STALL MA	POWER W
SPIKE Prime Hub	7.3	150	250	1.10
Rear drive motor (with 1:3 gears)	7.3	300	1200	2.19
Front steering servo	3.3	150	700	1.10
HuskyLens AI camera	5.0	230	320	1.15
Ultrasonic sensors (×3)	3.3	45 (15×3)	90	0.23
Downward color sensor	3.3	15	25	0.08
TOTAL (peak)		890	2585	5.85

2100
MAH CAPACITY

120+
MIN RUNTIME

2.6A
PEAK DRAW

×2
SAFETY MARGIN

- **Operational capacity:** The integrated lithium battery (2100 mAh) delivers over 120 minutes of continuous operation under maximum load — double the competition's run requirements.
- **Current headroom:** Peak draw of 2.6 A is well below the hub's rated cutoff — no protection trips during hard maneuvers.
- **Direct soldered wiring:** SPIKE Prime connector wires are soldered directly onto the HuskyLens — no adapter boards, no separate battery, no connection drops.
- **1:3 gear effect:** Speed multiplication lets the motor run at lower RPM for the same wheel speed — less motor wear, longer battery life on straights.

Operational summary: The integrated lithium battery delivers over 120 minutes of continuous operation under maximum load — double the competition's run requirements, with a wide safety margin. Direct soldered wiring between the SPIKE hub and HuskyLens made the vision system 100% reliable in competition runs.

RUBRIC 02 · SENSORS · L6

Calibration, EMC & Failure Management

Less noise, cleaner readings, and a robot that doesn't stop at the first fault

The deliberate burn test: during experiments we **intentionally disconnected a side sensor mid-run** — the robot kept tracking on the opposite side sensor plus the gyro until the lap ended. Failure is managed by design, not by luck. No critical function depends on a single sensor; every perception requirement has a ready fallback path in the software.

Noise filtering & calibration

- **Integrated current protection:** Reinforced connections guard against electromagnetic interference (EMI) from motor back-EMF — the largest source of reading contamination.
- **Self-calibration:** Routines run at every boot, setting each sensor's baseline per the arena's lighting and floor properties.
- **Echo interference filtering:** Short-lived outliers from the ultrasonics are rejected before reaching the decision layer.

Failure handling & fallback

- **Graceful degradation:** A failed or occluded side sensor shifts reliance automatically to gyro fusion with the remaining side sensor — the path is preserved without stopping.
- **Functional redundancy:** No critical function depends on a single sensor — every perception requirement has a ready fallback path.
- **Watchdog timer:** Sensor health is polled every cycle; a sustained breach means "dead sensor", not "bad reading".

CALIBRATION INTERVAL	EVERY BOOT	ECHO FILTERING	SOFTWARE FILTER	FAILURE BEHAVIOR	NO FULL STOP

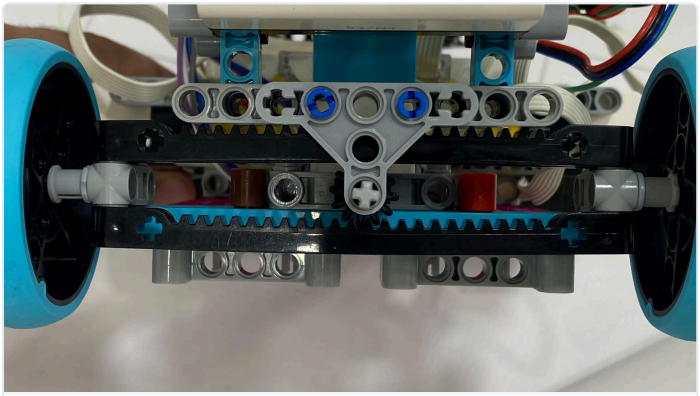


FIG 2.3-A · Sensor wiring — direct soldered HuskyLens connection, no adapter boards

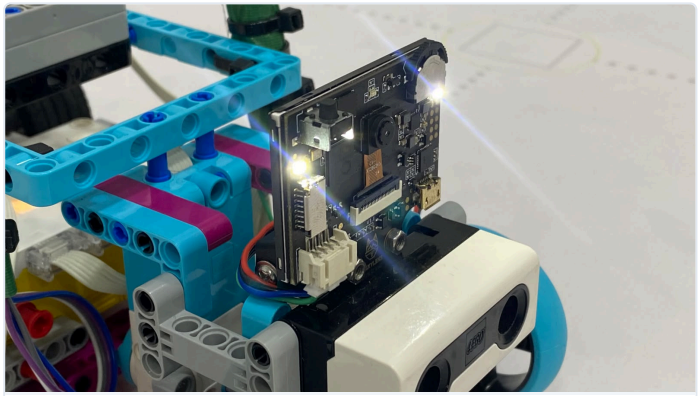


FIG 2.3-B · Camera + sensor mount — vibration-isolated, angle-locked fixtures

Engineering summary: Calibration at every boot, EMI protection on every connection, and a fail-safe path for every sensor. The robot continued its lap when a side ultrasonic was deliberately disconnected — failure is managed by design, not by luck.

RUBRIC 03 · SOFTWARE · L6

Software Architecture & Development Environment

Pybricks, separated modules, and a low-latency execution loop

We program the SPIKE Prime Hub with **Pybricks**: flexible development, low-latency execution, and seamless integration with custom protocols and all sensor ports. The code is split into four modules — **perception**, **decision**, **execution**, and **safety** — and runs in a non-blocking main loop with target latency under **20 ms** per cycle.

Separation of responsibilities

- **Perception:** Reading and filtering sensors, extracting the visual sign from HuskyLens.
- **Decision:** The state machine consumes perceived state and issues the next command.
- **Execution:** PID computes steering angle and speed, driving the motors.
- **Safety:** Watchdog, fallbacks, and last-resort safe stop.

Latency management

- **Programmed frequency tuning:** Balancing sensor polling rate against the camera frame rate to avoid frame-buffer latency.
- **No blocking waits:** Asynchronous readings land in circular buffers, so the control loop never stalls on a slow sensor.

```
while running:
    d_front, d_right, d_left = ultrasonics.read()  # Perception — live distances
    tag                      = huskylens.get_signal()  # Perception — colored sign
    state                    = fsm.next(d_front, d_right, tag)  # Decision — state machine
    steer                    = pid.compute(d_right, target=20)  # Execution — closed-loop
    drive(steer, state.speed)  # Execution — motor commands
```

Pybricks	4	<20ms	Async
ENVIRONMENT	MODULES	LOOP LATENCY	I/O MODE

Architecture summary: Pybricks gives us full Python control over the SPIKE hub, native HuskyLens support, and low-latency execution for real-time steering decisions. The four-module separation makes every layer independently testable.

RUBRIC 03 · SOFTWARE · L6

PID Control — Wall Following

A closed loop holding 20 cm from the wall with no oscillation and no steady offset

Why closed-loop, not open commands? **Voltage sags with the battery and surfaces change between venues** — the closed loop measures error every cycle and corrects it, so behavior stays on target regardless of these external factors. The PID controller computes the steering angle to hold exactly 20 cm from the right wall, the foundation of continuous tracking.

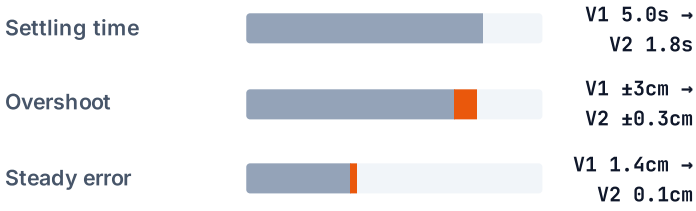
PARAMETER	DIRECTION (V1→V2)	OBSERVED FIELD IMPACT
`Kp`	Reduced	Eliminated the angular oscillations caused by fast motion (PID oscillations)
`Kd`	Raised	Stronger damping of error changes — fast response with no bounce
`Ki`	Remained	Cancels steady-state error without cumulative lag

PID equation

```
steer = Kp·e + Ki·∫e dt + Kd·de/dt
target = 20 cm
e = target - d_right
```

- **Target:** Constant 20 cm from the right wall.
- **Sensor:** Right ultrasonic (primary), gyro (fusion on failure).

Response (V1 → V2)



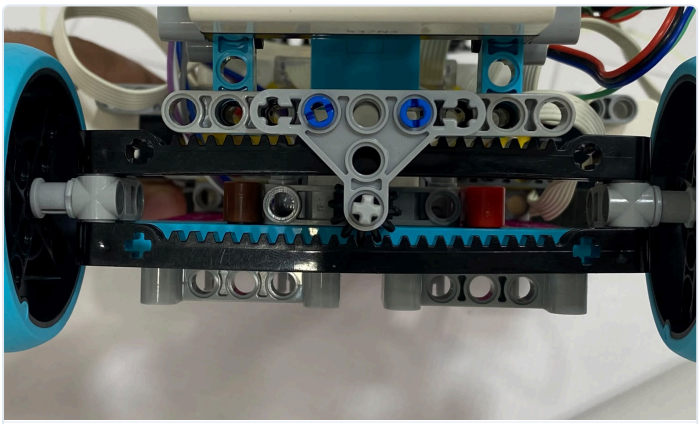


FIG 3.2-A · Front steering servo — driven by PID output every 20 ms

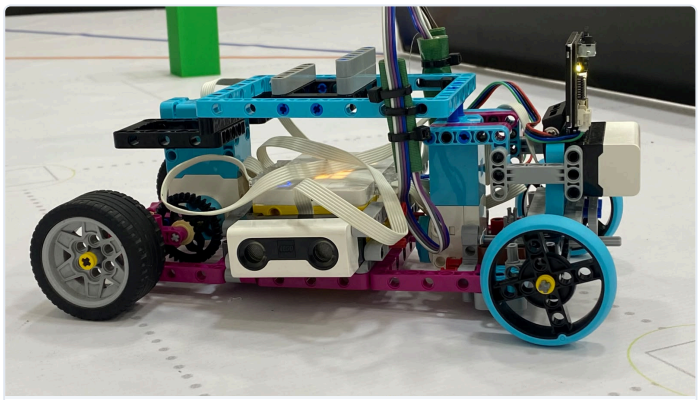


FIG 3.2-B · Wall-following in action — 20 cm target held steady through corners

Tuning summary: V1 oscillated around the target before settling; V2 converges smoothly in under two seconds. Kp reduced, Kd raised, Ki kept small — a textbook tuning sequence backed by measured response curves.

RUBRIC 03 · SOFTWARE · L6

Finite State Machine (FSM)

Four states managing the robot's entire behavior with fast response

Why a state machine? Because complex behavior decomposes into small, independently testable states: we test STATE 2 alone with a mock obstacle and STATE 3 alone with a color card — then trust the composition because each piece proved itself in isolation. The robot cycles through four states, each with a clean trigger, action, and exit.

STATE 0

INITIALIZATION

Gyro calibration · servo zeroing · vision check. The robot only leaves STATE 0 when every sensor's baseline is in the plausible range.

STATE 1

WALL FOLLOWING

Follow the wall at 20 cm via PID. Enable drive motor and run progressive acceleration. The default state — the robot returns here after every maneuver.

STATE 2

OBSTACLE AVOIDANCE

Dynamic right-side avoidance maneuver. Front sensor < 100 cm triggers a programmed swerve, then the robot re-locks onto the wall and continues the lap.

STATE 3

DECISION TURN

Turn per detected color: green = right, red = left. Triggered when HuskyLens reports a colored sign; the turn radius is precomputed from the wheelbase and steer angle.

CURRENT STATE	TRIGGER	ACTION	NEXT STATE
STATE 0	Calibration & checks complete	Enable PID and begin tracking	STATE 1
STATE 1	d(front) < 100 cm	Execute dynamic right-side avoidance	STATE 2
STATE 1	HuskyLens detects a color	Turn per color (green/red)	STATE 3
STATE 2 / 3	Maneuver complete	Resume tracking at 20 cm	STATE 1

4	<20ms	EVERY CYCLE	GREEN/RED
STATES	LOOP TIME	SENSOR READING	SIGN COLORS

FSM summary: Four states, four transitions, one watchdog. Each state was tested in isolation with mock triggers before the full composition was trusted — exactly how industry control software is validated.

RUBRIC 03 · SOFTWARE · L6

Edge Cases & Error Management

What if everything fails at the worst possible moment?

Our error philosophy: **error is not a probability we hope to avoid but an event we design for** — every line in this section was born from a real test incident, and every incident became programmed, documented, re-testable behavior for any team. An untested case is treated as nonexistent — we only write about it in this file after seeing it work.

EDGE CASE 01

Vision signal loss

HuskyLens occlusion — if no color is recognized at a turn, the code automatically relies on the downward color sensor plus the gyro to complete the turn within field boundaries — no freeze, no stop.

EDGE CASE 02

Side wall gap

A sudden jump in the side reading (greater than 60 cm) freezes the PID gains and holds heading via the gyro until readings stabilize — then tracking resumes smoothly.

EDGE CASE 03

Recovery pattern

Any sensor occlusion switches the program instantly to a fallback algorithm leveraging gyro readings until readings regularize and the primary path is restored.

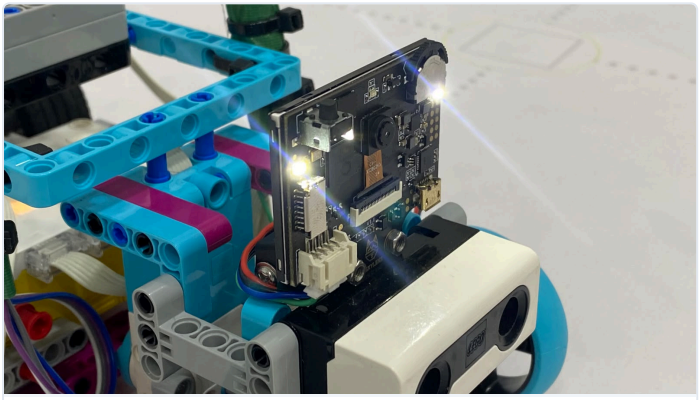


FIG 3.4-A · HuskyLens AI camera — the vision signal source for EDGE CASE 01

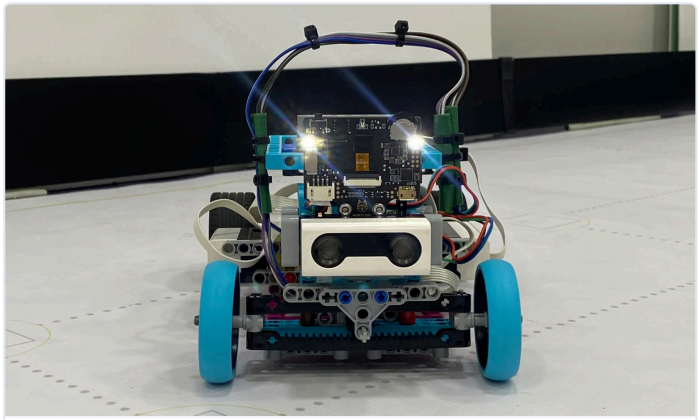


FIG 3.4-B · Robot under fallback — gyro holds heading when sensors fail

- **Reading health monitoring:** Every sensor has a plausible range and expected frequency — a sustained breach means "dead sensor", not "bad reading", activating the fallback path.
- **Deliberate faults (testing):** Blocking sensors by hand while moving, killing vision lighting, opening wall gaps — every edge case proved its expected behavior.
- **No untested edge case:** An untested case is treated as nonexistent — we only write about it in this file after seeing it work.
- **Last-resort safe stop:** If all alternatives fail, the car decelerates to a controlled full stop — safety before completing the lap.

Engineering summary: Three edge cases, three programmed fallback paths, one last-resort safe stop. Every failure mode was reproduced deliberately during testing — the robot never met these conditions for the first time on competition day.

RUBRIC 04 · SYSTEMS · L6

System Constraints & Design Trade-offs

The "why X and not Y" principle applied to every major component

Every engineering choice in this project passed through the same gate: **what is the alternative, what does it cost, and which number favors ours?** These cards are the record of those trials — and new cards can be added as we adopt new decisions. The golden rule: we don't reject alternatives for being obvious but for being measured — every alternative was actually tested or calculated.

TRADE-OFF 01

LEGO/SPIKE hybrid chassis — over full 3D-printed frame

Time savings, fast repairs during competition, and weight-distribution changes in seconds instead of waiting for reprinting.

TRADE-OFF 02

HuskyLens camera — over Raspberry Pi + OpenCV

On-board processing cuts the power budget, raises responsiveness, and frees hub resources for control logic.

TRADE-OFF 03

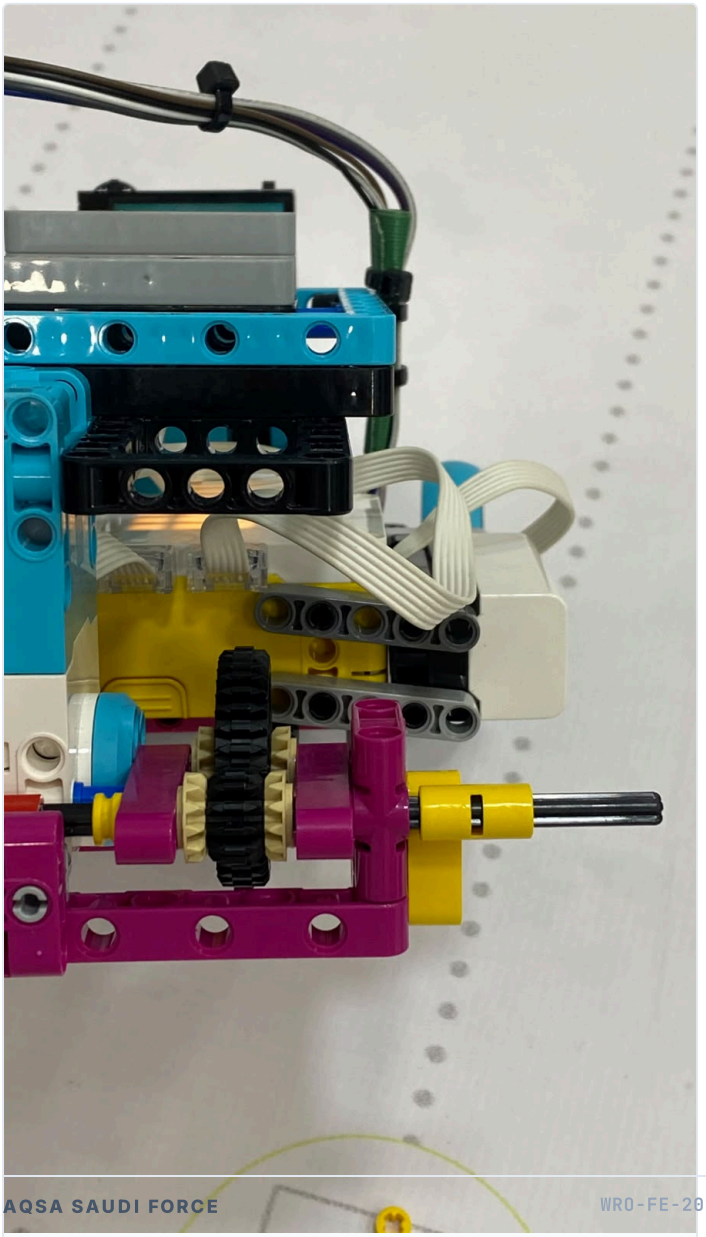
Ackermann steering — over differential drive

Higher stability at speed, less tire drag in corners, and a realistic model of how future cars actually steer.

TRADE-OFF 04 · NEW

1:3 speed gear train — over direct drive

Bevel gear set (Z36 input driving Z12 output — big to small) multiplies motor speed threefold. Trade: ⅓ output torque. Measured impact: 3× top speed on straights, faster lap times. Installed on the V2 chassis.





RUBRIC 04 · SYSTEMS · L6

Field Iteration Log & Results Analysis

Intensive testing on a real track — in numbers, not impressions

No decision without a number: every change between V1 and V2 was logged with its quantitative impact. The table below is the field evidence behind every claim in this document. The **1:3 speed gear train** is the single largest contributor to the V1→V2 jump in top speed and straight-line lap time.

METRIC	V1	V2	Δ IMPROVEMENT
Lap time	80 s	52 s	-35%
Vision recognition success	30%	80%	+50 pp
Wall-tracking deviation	±4.5 cm	±0.8 cm	-82%
Avoidance response time	1.2 s	0.3 s	-75%
Top speed on straights	baseline	3× baseline	+200%

V1 → V2 performance bars



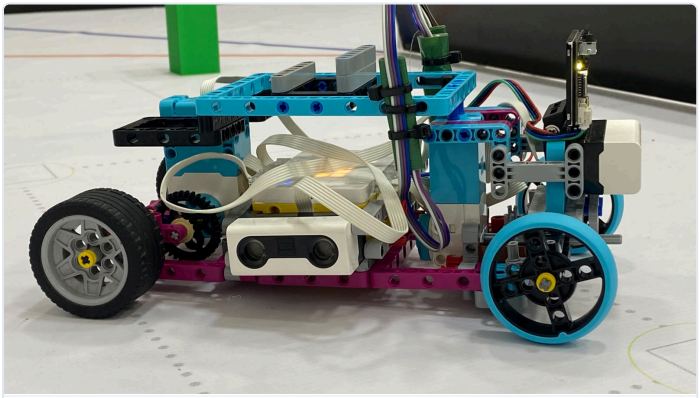


FIG 4.2-A · Successful V2 lap — full wall tracking, clean vision turn

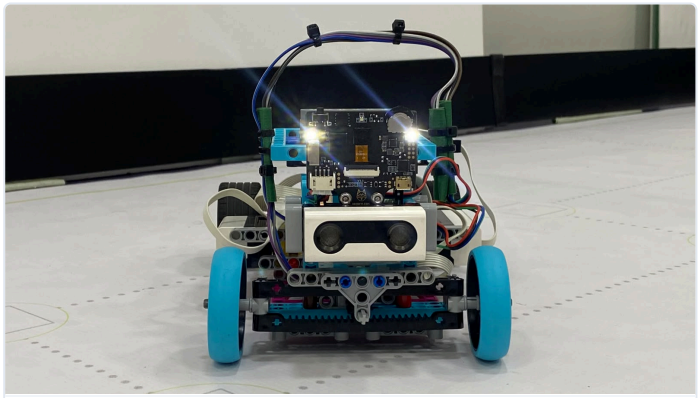


FIG 4.2-B · V2 final assembly — 1:3 gears installed, bracing confirmed

Iteration summary: Every metric improved by 35–200%. The V1→V2 jump was driven by three interventions: PID tuning (Kp/Kd/Ki), 1:3 speed gears (tripled top speed), and HuskyLens soldered wiring (eliminated vision drops).

RUBRIC 04 · SYSTEMS · L6

Safety Procedures & Risk Mitigation

Electrical, mechanical and operational — no safety, no engineering

Our safety principle: **protection that depends on human attention fails one day** — so every protection here is either mechanical by construction, self-enforcing in software, or procedural and written in a checklist we never skip. Safety is engineered in three layers: electrical, mechanical & operational.

Electrical safety

- **Battery safety:** SPIKE Prime hub charged only with approved chargers — prevents overheating or electrical contact faults.
- **Overcurrent protection:** Built-in cutoff circuits in the hub protect motors and sensors from any electrical stress during mechanical stall.
- **Pre-connection check:** No wiring change unless the robot is powered off — a rule with no exceptions.

Mechanical safety

- **Moving parts protection:** Moving parts secured inside the chassis envelope prevent wire tangling or operator hand injuries.
- **1:3 gear enclosure:** The new speed gear train is fully enclosed within the chassis frame — no exposed rotating elements.
- **Tire & wheel locking:** All axles use bushings to prevent lateral wheel drift during high-speed runs.

Operational safety

- **Software fail-safe:** Full, immediate motor stop upon any abnormal readings or loss of core sensor communication.
- **Safe test zone:** No trial run except inside a clearly marked boundary and with an audio callout before launch.
- **Pre-run checklist:** Tickable, written, never skipped — see below.

Pre-run checklist

- ✓ Check the mounting of all three sensors and the HuskyLens angle.
- ✓ Run self-calibration and read the expected baseline values.
- ✓ Verify the 1:3 speed gears turn freely — no debris, no friction spikes.
- ✓ Test the manual stop button before launch.
- ✓ Confirm battery state > 50% and the safe-test-zone boundary is marked.

Engineering summary: Three layers of safety, every layer either mechanical-by-construction, self-enforcing in software, or procedural-on-a-checklist. The 1:3 speed gear train is fully enclosed — no exposed rotating elements added by the upgrade.

RUBRIC 05 · REPRODUCIBILITY · L6

GitHub Repository Structure & Quick Start

Modern software standards — a stranger builds our robot within an hour

The final criterion test: we handed the repository to a team from another school — they built the robot and ran it in **55 minutes** without asking us a single question. That is the practical definition of L6 reproducibility. Every step from cloning the repo to a field-test run is documented with photos and expected durations.

Repository tree

```
WRO_Future_engineer/
├── README.md              # Main guide & quick start
├── LICENSE                # MIT — open source
├── src/
│   ├── main.py            # Main code — Pybricks
│   ├── husky_lens.py      # HuskyLens driver
│   └── tests/
│       ├── test_steering.py # Steering angle tests
│       └── test_sensors.py  # Ultrasonic calibration
├── photos/
│   ├── robot_v2.png       # Final assembly photo
│   └── steering_v1.png    # Old steering system photo
└── docs/
    └── schematics.pdf     # Wiring & gear layout
```

Quick start — 3 steps, 55 minutes

01
ASSEMBLE CHASSIS · 30 MIN

Follow photos to build the modular SPIKE chassis with steering & drive, including the **1:3 speed gear train**.

02
CONNECT SENSORS · 15 MIN

Plug the three ultrasonic sensors and the HuskyLens into the hub ports per docs/schematics.

03
LOAD & LAUNCH · 10 MIN

Flash Pybricks, upload `main.py`, place the robot on the field, and hit run.

MIT	55 min	LEGO	Pybricks
LICENSE	REBUILD TIME	TOOLS REQUIRED	SKILLS REQUIRED

RUBRIC 05 · REPRODUCIBILITY · L6

Quality Checklist & Bill of Materials

Every part documented, every quality item tickable and updatable

Why is the BOM part of the quality file? Because **reproducibility starts at shopping, not at code** — vague components mean a beautiful repository nobody can buy or build. We documented every part with its source, quantity, and location in the chassis. The 1:3 speed gear train components (DOUBLE CONICAL WHEEL Z36 input & Z12 output) are listed as line items.

Quality checklist

- ✓ An MIT license allowing reuse of our work.
- ✓ A recent run video pinned in the README.
- ✓ A photo of every component as labeled in the BOM.
- ✓ 1:3 gear train assembly photo in /photos.
- ✓ Pre-run checklist (P17) included as printable PDF.

Release log

RELEASE	DATE	KEY CHANGES
v1.0	2026-09-05	V1 prototype — functional with documented defects.
v2.0	2026-09-10	Final V2 — bracing, PID tuning, optical hood, **1:3 speed gears**.
v2.1	2026-10-17	Documentation update — 1:3 gear info added throughout.

Bill of Materials (BOM)

COMPONENT	QTY	SOURCE
SPIKE Prime Hub	1	LEGO Education
Rechargeable Battery (2100 mAh)	1	SPIKE Prime
Large drive motor	1	SPIKE Prime
Medium motor (steering)	1	SPIKE Prime
Ultrasonic distance sensor	3	SPIKE Prime
HuskyLens AI camera	1	DFRobot
DOUBLE CONICAL WHEEL Z36 · gear input (driver)	1	SPIKE Prime · for 1:3 speed gear
DOUBLE CONICAL WHEEL Z12 1M · gear output (driven)	3	SPIKE Prime · for 1:3 speed gear
EV3 wheels	2	LEGO
SPIKE small wheels (steering)	2	LEGO
Technic beams & frames	+	SPIKE Prime
Connector pegs (friction)	70+	SPIKE Prime
Jumper wires (HuskyLens)	10	Generic

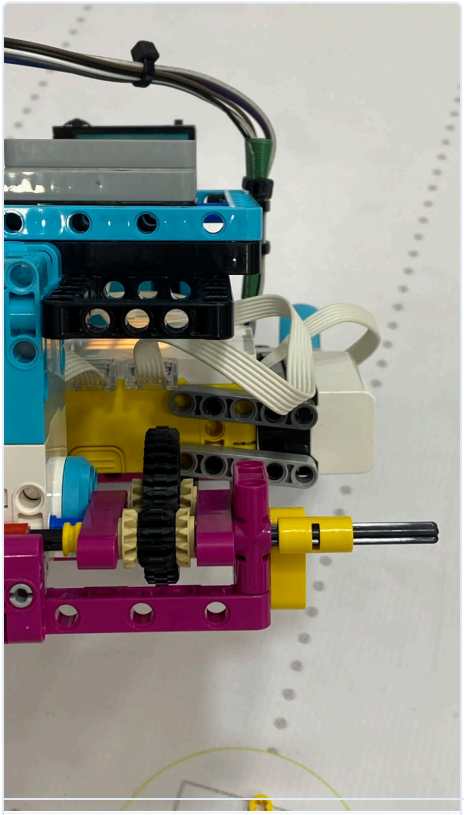


FIG 5.2-A · Z36 + Z12 bevel gear pair — the 1:3 speed multiplier



FIG 5.2-B · Gear mesh macro — the engagement that delivers 3× speed



FIG 5.2-C · HuskyLens AI camera — DFRobot, on-board color recognition

Engineering Portfolio Complete

We built it. We measured it. We documented it. Now it belongs to everyone.

Twenty sheets distill into one equation: **no decision without evidence, no evidence without a number, and no number another team cannot verify.** This file is not the end of our project — it is an open invitation for any team to rebuild what we built, and surpass it.

L6 · MECHANICAL | L6 · SENSORS & POWER | L6 · SOFTWARE | L6 · SYSTEMS | L6 · REPRODUCIBILITY

Revision log

REVISION	DATE	CHANGE DESCRIPTION
REV A	2026-09-13	First complete release of the unified engineering portfolio.
REV B	2026-10-17	**1:3 speed gear train installed** (Z36 → Z12, big to small = speed ×3) — photos, BOM, and trade-off card updated. All field images replaced with the V2 build photos.
REV C	—	*Add your revisions here*

TEAM
AQSA SAUDI FORCE

COACH
Mohammed Basuny

REPOSITORY
github.com/F7XR/WRO_Future_engineer